



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.EE.2C

# Evaluate Expressions

Lesson 6-2 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

**x**

stands for a number

*In  $3x + 5$ ,  $x$  can be any number — if  $x = 2$ , the expression equals 11; if  $x = 10$ , it equals 35*

### Variable

Write the definition:

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**$3x + 5$**

no equal sign

*$2n + 7$  is an expression: the  $2n$  means '2 times some number' and the  $+ 7$  means 'then add 7'*

### Expression

Write the definition:

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**$3x + 5$**

no equal sign

*Evaluate  $2n + 7$  when  $n = 4$ : replace  $n$  with 4  $\rightarrow$   
 $2(4) + 7 = 8 + 7 = 15$*

### Evaluate

Write the definition:

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**$x \rightarrow 5$**

put a number in for  $x$

*Substitute 3 for  $x$  in  $5x$ : replace  $x$  with 3  $\rightarrow 5(3) = 15$*

### Substitute

Write the definition:

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$$4x$$

coefficient = 4

*In  $7m$ , the coefficient 7 tells you to multiply  $m$  by 7  
— if  $m = 3$ , then  $7m = 21$*

### Coefficient

Write the definition:

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$$3x + 2x = 5x$$

same variable → combine

*$4x$  and  $2x$  are like terms (both  $x^1$ );  $4x$  and  $4y$  are  
NOT (different variables)*

### Like terms

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can evaluate algebraic expressions by substituting values for the variables.**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Variable

Expression

Evaluate

Substitute

Coefficient

Like terms



*Tap any word to see what it means and a picture.*

**1** A letter that stands for an unknown number, like  $x$ , is a .

**2** A group of numbers, variables, and operations with no equal sign is an .

**3** To find the value of an expression for given values is to  it.

**4** To replace a variable with a number is to .

**5** A number multiplied by a variable, like the 4 in  $4y$ , is a .

**6** Terms with the same variable part, like  $2x$  and  $6x$ , are .



### Watch & Try — Worked Examples

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** Evaluate  $3x + 7$  when  $x = 4$ .

- A. 19
- B. 34
- C. 12
- D. 14

**Step 1**  $3(4) + 7 = 12 + 7 = 19$ .

 **Answer:** A. 19

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** Evaluate  $20 - 2n$  when  $n = 6$ .

- A. 8
- B. 14
- C. 32
- D. 12

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** Which expression means '5 less than twice a number  $n$ '?

A.  $2n - 5$

B.  $5 - 2n$

C.  $2(n - 5)$

D.  $2 - 5n$

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Track 2 — The volume mix is  $2(n + 3)$ , where  $n$  is the number of monitors. Evaluate  $2(n + 3)$  when  $n = 5$ .**

- A. 16
- B. 13
- C. 20
- D. 26

Show your work:

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**2. Track 3 — The speaker power is  $x^2$ , where  $x$  is the gain setting. Evaluate  $x^2$  when  $x = 6$ .**

- A. 36
- B. 12
- C. 18
- D. 30

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A gym membership costs \$25 per month plus a \$40 signup fee. Write an expression using  $m$  for months, then evaluate it for 6 months and 12 months. Explain what each part of the expression represents.**

*Sentence starter: The expression is \_\_\_\_\_. For 6 months: \_\_\_\_\_ = \_\_\_\_\_. For 12 months: \_\_\_\_\_ = \_\_\_\_\_. The \_\_\_\_\_ represents the \_\_\_\_\_ and the \_\_\_\_\_ represents the \_\_\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**Evaluate  $5(n - 3) + 2$  when  $n = 7$ .**

- A. 22
- B. 32
- C. 17
- D. 36

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Variable
2. **Notes 2:** Expression
3. **Notes 3:** Evaluate
4. **Notes 4:** Substitute
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A.  $8 - 20 - 2(6) = 20 - 12 = 8$ .
8. **You Try (notes):** A.  $2n - 5$  — *Twice  $n$  is  $2n$ . Five less than that is  $2n - 5$ .*
9. **Try It 1:** A.  $16$  — *Substitute  $n = 5$ :  $2(5 + 3)$ . Do the parentheses first:  $5 + 3 = 8$ . Then multiply:  $2 \times 8 = 16$ .*
10. **Try It 2:** A.  $36$  —  *$x^2$  means  $x$  times itself. Substitute  $x = 6$ :  $6^2 = 6 \times 6 = 36$ .*
11. **Exit Ticket:** A.  $22 - 5(7 - 3) + 2 = 5(4) + 2 = 20 + 2 = 22$ .